Phase 2.5 — Practical Assignments (Week 2)

Week 2 — Processes & Signals

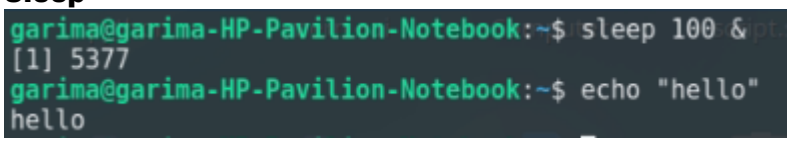
🎯 Assignment 1: Process Investigation

1. Start multiple background processes:
   - sleep
   - yes > /dev/null

2. Use:
   - ps
   - top
   - pstree
   - lsof

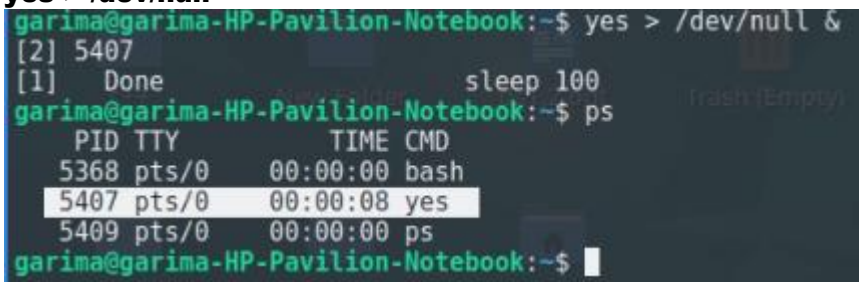
Solutions: -

1. **Background processes**:
   - Sometimes we want a program to run **in the background** so we can keep using the terminal.
     (a) **Sleep**



This command will wait for 100 sec.& will make it run in background. We can still use the terminal

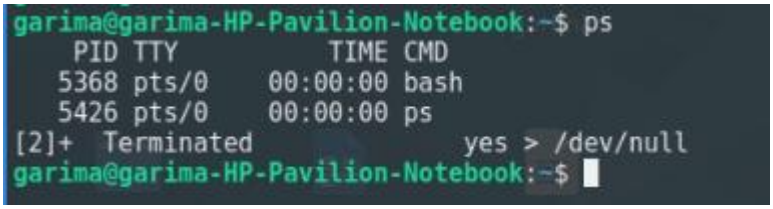
(b) **yes > /dev/null**



yes, keeps printing “y” again and again. /dev/null throws away all output (so screen doesn’t fill up). Runs continuously in background

2. **Basic commands:**

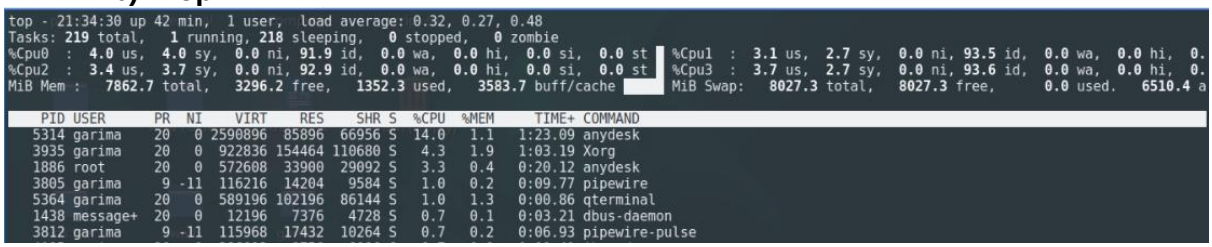
a) Ps:



shows acurrent running processes

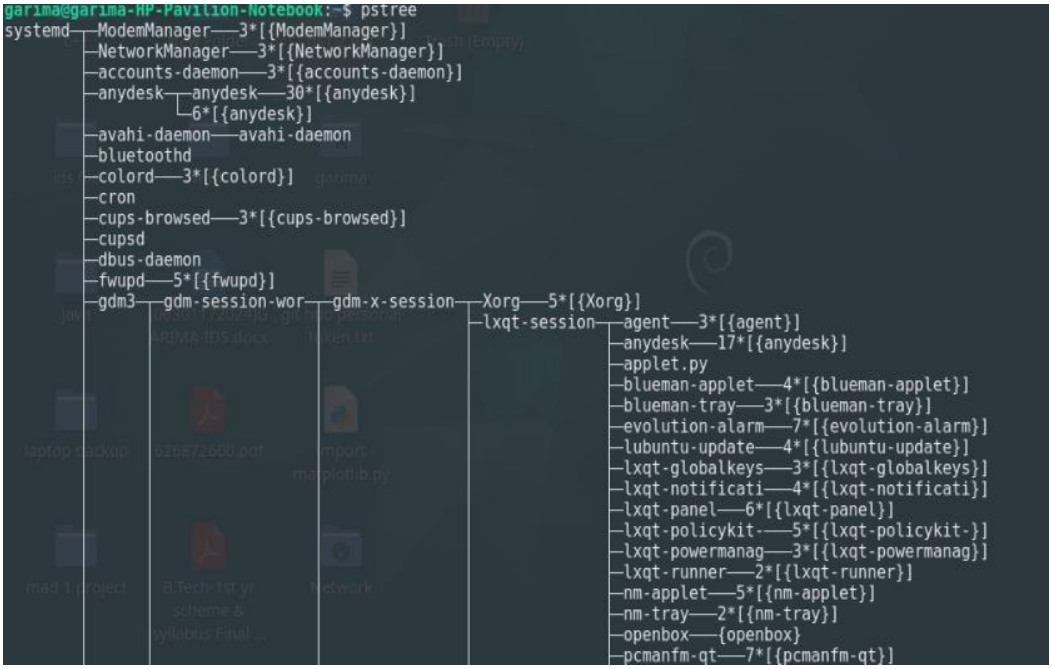
shows all processes with full details

b) Top:



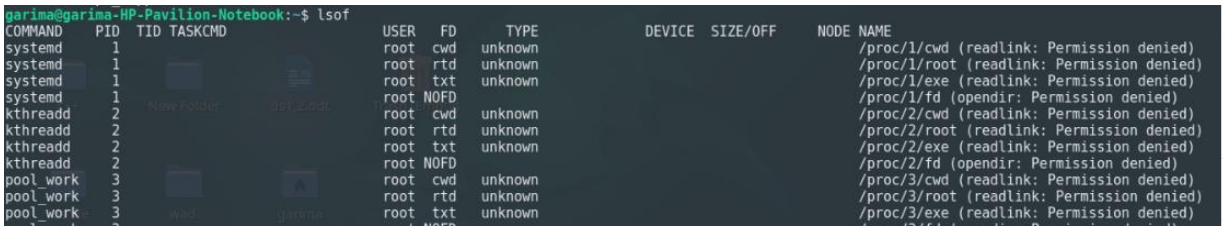
Shows live processes (updates continuously). You can see CPU and memory usage. Press `q` to exit

c) Pstree:



shows processes in a tree format. Helps understand which process started which

d) Lsof:



Shows which processes opened which all files. Useful to check files .

🎯 Assignment 2: Signal Lab

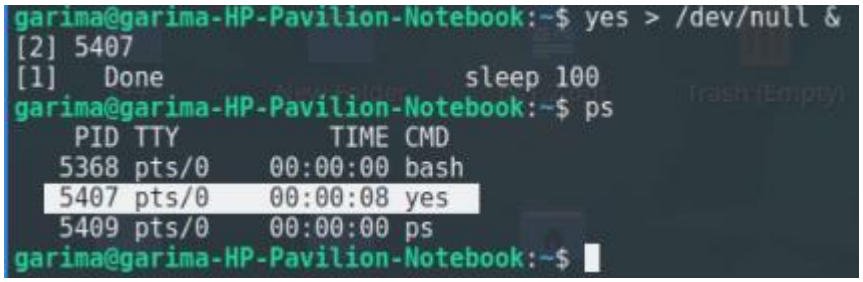
1. Start a long-running process.

2. Send:

   - SIGINT
   - SIGTSTP
   - SIGCONT
   - SIGKILL

Solutions:

1.long-running process:



Yes > /dev/null will run till its gets manual terminated or is controlled by a function or a script.

2.Signals and Their Behavior:

Signal	How to Send	What it Does	Behavior Type
SIGINT	`Ctrl + C`	Stops the process	Normal stop
SIGTSTP	`Ctrl + Z`	Pauses the process	Temporary stop
SIGCONT	`bg` / `kill -CONT PID`	Resumes paused process	Continue execution
SIGKILL	`kill -9 PID`	Forcefully kills process	Immediate force stop

Easy way to remember

- **SIGINT** → Stop
- **SIGTSTP** → Pause
- **SIGCONT** → Resume
- **SIGKILL** → Kill (forcefully)

🎯 Assignment 3: PATH Hijack Experiment

1. Create a fake command in a custom directory.

2. Modify PATH.

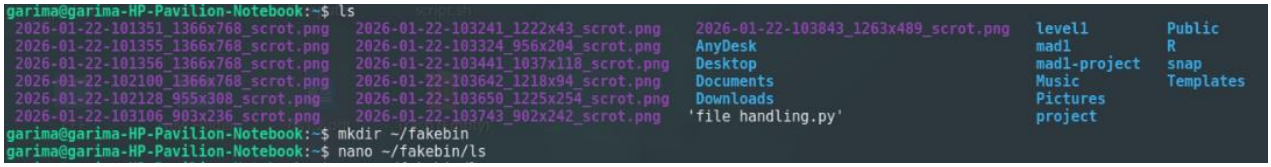
3. Override system command behavior.

Solution:

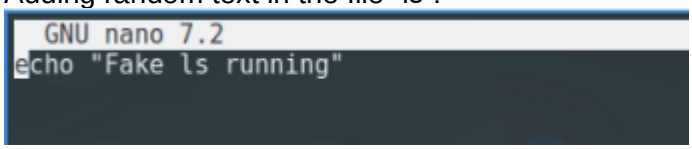
1. Create a fake command

mkdir ~/fakebin

nano ~/fakebin/ls



Adding random text in the file “ls”:

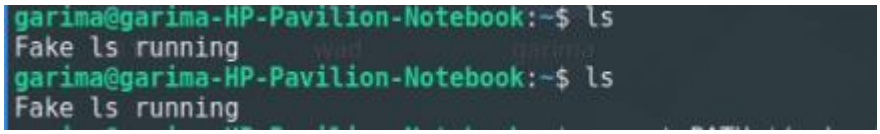


Make it executable:

2.Modify PATH

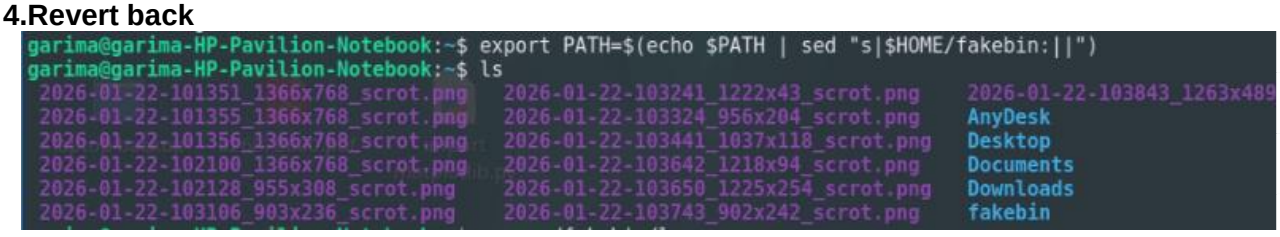
This puts the fake folder **before system folders.**

3.Override system command



Instead of real `ls`, the fake one runs

4.Revert back



or simply restart terminal

5. Why this is dangerous

- System checks commands in **PATH order**
- If fake command comes first → it replaces real one
- Can be misused to:
    - Steal passwords
    - Run harmful scripts
    - Trick users without them knowing